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Topic 4: The Carbon Cycle & Energy Security

4.1 The Carbon Cycle as a System

Carbon cycle: The biogeochemical cycle by which carbon is exchanged between the Earth's atmosphere, biosphere, hydrosphere, pedosphere (soils), and lithosphere. It operates as a closed system at the global scale (no carbon enters or leaves Earth, though energy does). Carbon is the building block of all organic molecules and a key component of greenhouse gases (CO_2 and CH_4), making the carbon cycle central to climate regulation.

Carbon Stores

Store	Approximate Size (Gt C)	Timescale	Key Details
Lithosphere (rocks, fossil fuels)	~100,000,000	Millions of years (slow cycle)	By far the largest store. Limestone ($CaCO_3$), chalk, coal, oil, gas. Carbon locked in sedimentary rocks cycles over geological time through weathering, sedimentation, and volcanism.
Oceans	~38,000	Hundreds to thousands of years	Deep ocean is the main store. Surface ocean exchanges rapidly with atmosphere. Dissolved inorganic carbon (DIC), dissolved organic carbon (DOC), marine organisms. Ocean absorbs ~25% of annual anthropogenic CO_2.
Soils and peat	~2,500	Decades to millennia	Humus, decomposed organic matter, peat. Top 1 m of soil contains more carbon than atmosphere + vegetation combined. Peatlands store ~600 Gt C despite covering only 3% of land.

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Permafrost	~1,700	Thousands of years (thawing)	Frozen organic matter in Arctic soils. Contains ~2x atmospheric carbon. Thawing under climate change releases CO_2 and CH_4 — a major positive feedback risk.
Atmosphere	~870 (at ~420 ppm)	~5 years per molecule	CO_2 (~420 ppm, up from 280 ppm pre-industrial), CH_4 (~1,920 ppb), and other trace gases. The store is small but its size critically determines global temperature.
Biosphere (living organisms)	~560	Years to decades	Carbon in plant biomass (90%), animal biomass, and microorganisms. Tropical forests hold ~55% of terrestrial biomass carbon. The Amazon alone stores ~150–200 Gt C.
Fossil fuels (recoverable)	~4,000	Millions of years to form	Coal (~65%), oil (~15%), gas (~20%). Formed from ancient organic matter buried and compressed. Non-renewable on human timescales. Burning transfers carbon from slow geological cycle to fast atmospheric cycle.

Key Carbon Fluxes

- **Photosynthesis** — ~120 Gt C/yr absorbed from atmosphere by terrestrial plants: $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$. The largest natural flux from atmosphere to biosphere. Rate depends on light, CO_2 concentration, temperature, and water availability.
- **Respiration** — ~119 Gt C/yr returned to atmosphere. All living organisms release CO_2 through cellular respiration. Nearly balances photosynthesis, making the biosphere approximately carbon-neutral over geological time.
- **Decomposition** — bacteria and fungi break down dead organic matter. Aerobic decomposition releases CO_2; anaerobic decomposition (waterlogged conditions) releases CH_4 (methane, ~80x more potent than CO_2 over 20 years). Rate depends on temperature, moisture, oxygen, and material type.
- **Fossil fuel combustion** — ~9.5 Gt C/yr (2023). The primary human disruption. Transfers carbon from geological stores (accumulated over millions of years) to the atmosphere in decades. Coal, oil, gas burning for energy, transport, industry, and heating.

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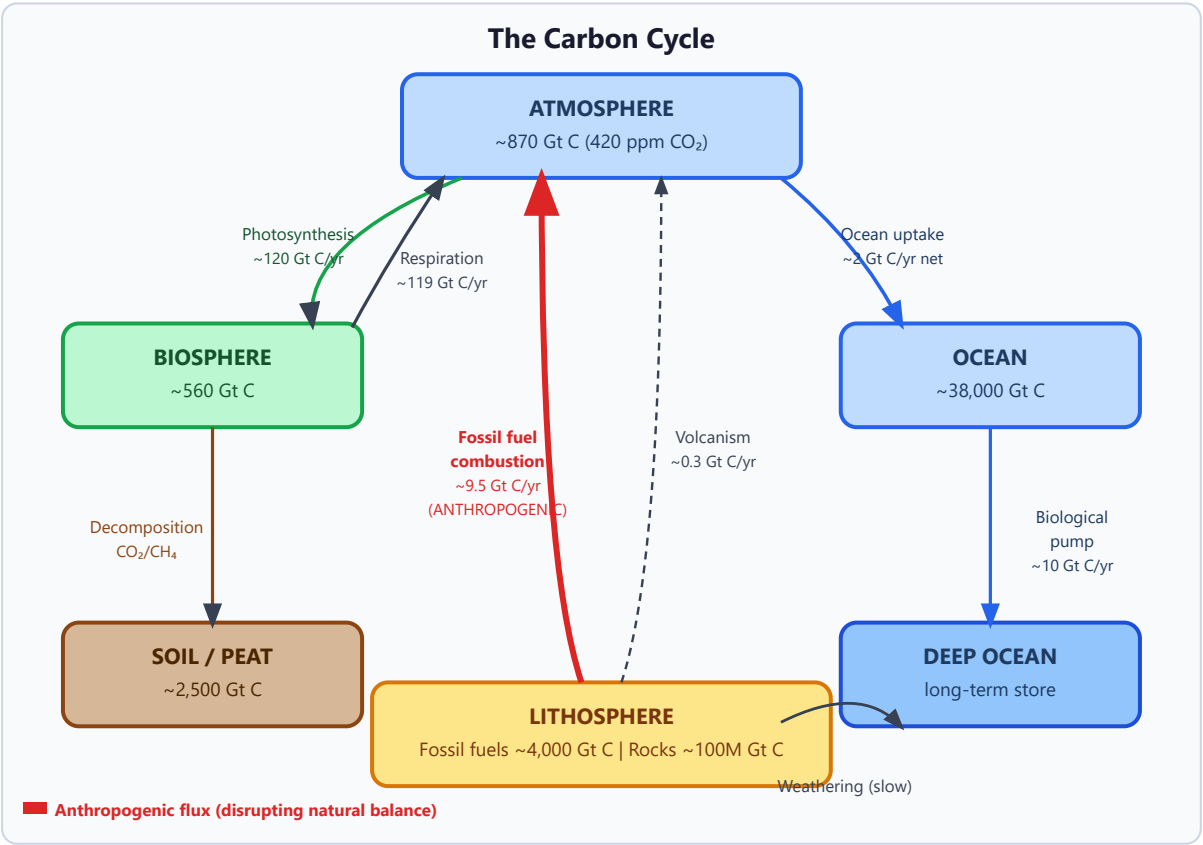
- **Deforestation and land use change** — ~1.5 Gt C/yr. Burning and decomposition of cleared vegetation releases stored carbon. The lost forest can no longer sequester carbon. ~95% from tropical deforestation.
- **Ocean-atmosphere exchange** — ~92 Gt C/yr absorbed by oceans, ~90 Gt C/yr released. Net ocean absorption: ~2 Gt C/yr. Cold polar waters absorb more CO_2 (solubility pump); warm tropical waters release it. The ocean has absorbed ~30% of cumulative anthropogenic CO_2 since 1750.
- **Biological pump** — marine phytoplankton photosynthesise in the sunlit surface layer, incorporating carbon. When they die, organic matter sinks to deep ocean (marine snow), transferring carbon to long-term storage. This pump removes ~10 Gt C/yr from the surface ocean.
- **Chemical weathering** — CO_2 dissolved in rainwater forms carbonic acid (H_2CO_3), which dissolves silicate and carbonate rocks: $CaCO_3 + H_2CO_3 \rightarrow Ca^{2+} + 2HCO_3^-$. Transfers atmospheric carbon to dissolved form in rivers and eventually ocean sediments. Very slow (~0.3 Gt C/yr) but important over geological time.
- **Volcanic outgassing** — ~0.3 Gt C/yr released from mantle to atmosphere. Tiny compared to anthropogenic emissions but critical for long-term carbon cycle balance over millions of years.



Fast vs Slow Carbon Cycle

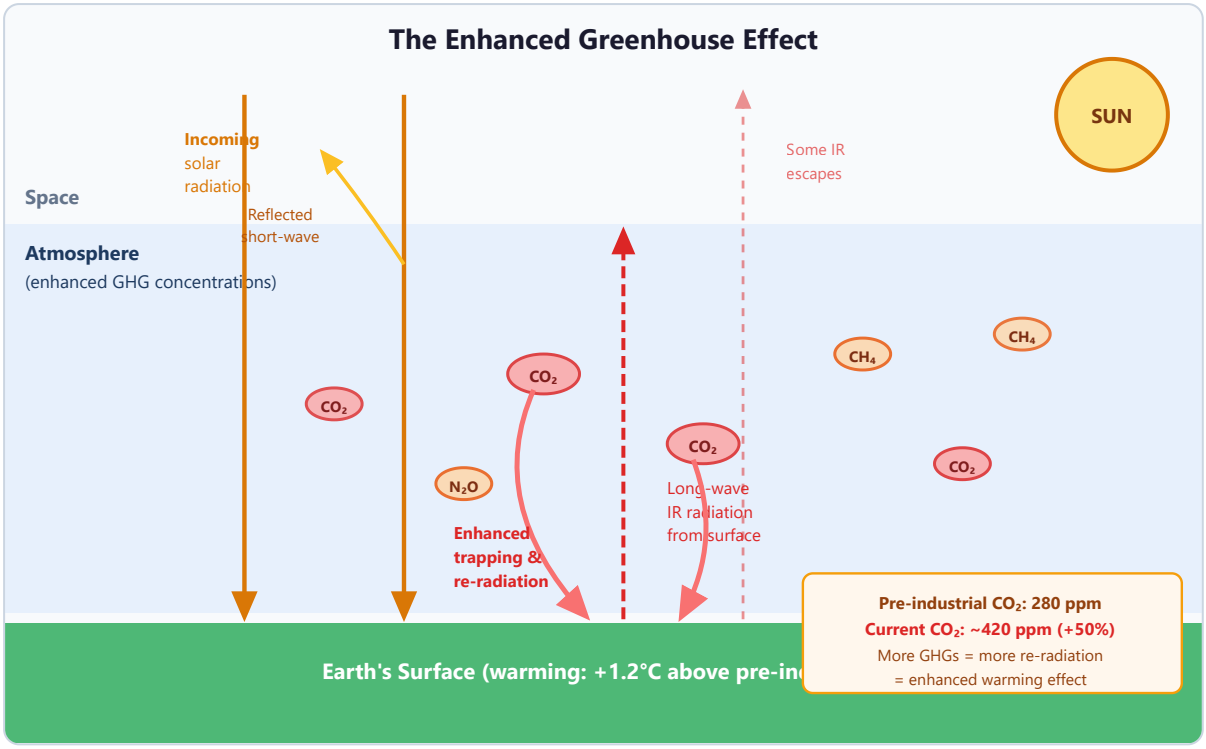
Fast Carbon Cycle	Slow Carbon Cycle
Operates over years to decades	Operates over millions of years
Biological processes: photosynthesis, respiration, decomposition	Geological processes: weathering, sedimentation, lithification, volcanism, fossil fuel formation
Moves ~120 Gt C/yr between atmosphere and biosphere	Moves ~0.1–0.3 Gt C/yr between atmosphere and lithosphere
Short-term climate regulation	Long-term climate regulation (thermostat over millions of years)
Human impact: deforestation, agriculture alter fast cycle	Human impact: fossil fuel burning transfers carbon from slow cycle to fast cycle at unprecedented rate

4.2 Human Disruption of the Carbon Cycle

The Enhanced Greenhouse Effect

The natural greenhouse effect keeps Earth ~33°C warmer than it would otherwise be (global average ~15°C instead of −18°C). Human activities have enhanced this by increasing atmospheric concentrations of greenhouse gases:

- **CO_2** — risen from ~280 ppm (pre-industrial) to ~420 ppm (2024). A 50% increase, unprecedented in at least 800,000 years (ice core evidence). Primary source: fossil fuel combustion (~75%) and land use change (~25%).
- **CH_4 (methane)** — risen from ~722 ppb to ~1,920 ppb. Sources: agriculture (rice paddies, ruminant livestock), fossil fuel extraction (gas flaring, coal mining), landfill, wetlands, permafrost thaw.
- **N_2O (nitrous oxide)** — risen ~23%. Sources: agricultural fertilisers, livestock waste, industrial processes. ~300x more potent than CO_2.



Consequences of Rising Atmospheric Carbon

- **Global warming** — global mean temperature has risen ~1.2°C above pre-industrial levels (2023 was the warmest year on record). IPCC projections: 1.5–4.5°C by 2100 depending on emissions pathway.
- **Ocean acidification** — CO_2 dissolving in seawater forms carbonic acid. Ocean pH has fallen from 8.2 to 8.1 since 1750 (26% increase in acidity). Threatens coral reefs (bleaching), shell-forming organisms (pteropods, oysters), and marine food chains. The Great Barrier Reef experienced mass bleaching events in 2016, 2017, 2020, 2022, and 2024.
- **Cryosphere loss** — accelerated melting of glaciers, ice caps, and ice sheets. Arctic sea ice has declined ~40% since 1979. Greenland Ice Sheet loses ~280 Gt/yr; Antarctic Ice Sheet ~150 Gt/yr. Contributes to sea-level rise.
- **Sea-level rise** — ~20 cm since 1900, currently ~3.6 mm/yr (accelerating). Threatens low-lying coastal areas, small island developing states, and delta regions. IPCC: 0.43–1.1 m rise by 2100.
- **Extreme weather** — more frequent and intense heatwaves, droughts, heavy rainfall events, and tropical cyclone intensification.
- **Ecosystem shifts** — species range migration poleward/upward, phenological mismatches, coral bleaching, forest dieback, permafrost ecosystem collapse.

Case Study: Tropical Rainforests — Carbon Store Under Threat (Amazon)

The Amazon rainforest (5.5 million km^2, 9 countries, ~60% in Brazil) is the world's largest tropical forest and stores an estimated 150–200 Gt C in biomass and soils. It acts as a significant carbon sink, absorbing ~0.5 Gt C/yr through photosynthesis. **Threats:** Deforestation peaked at ~27,000 km^2/yr (2004), fell to ~5,000 km^2/yr under enforcement, but fluctuated with political changes. Drivers: cattle ranching (80% of deforested land), soy farming, logging, mining, road construction (BR-163 highway opened vast areas). **Carbon impacts:** deforestation releases stored carbon immediately (burning) and over decades (decomposition). Research (2021 Nature study) found that parts of the eastern Amazon have become a net carbon source (emitting more CO_2 than they absorb) due to combined effects of deforestation and increasing drought/fire frequency. **Tipping point risk:** the Amazon generates ~50% of its own rainfall through transpiration (moisture recycling). Scientists estimate that at 20–25% forest loss (~17% already cleared), the system could cross a tipping point where reduced rainfall causes irreversible dieback, converting forest to savanna and releasing 50+ Gt C. **Significance:** losing the Amazon as a carbon sink would make Paris Agreement targets virtually impossible to achieve.

Case Study: Arctic Permafrost — The Carbon Feedback Time Bomb

Permafrost (permanently frozen ground) covers ~23 million km^2 of the Northern Hemisphere (23% of land surface) and contains ~1,700 Gt C — roughly twice the atmospheric carbon store. As the Arctic warms at 2–4 times the global average rate (Arctic amplification due to ice-albedo feedback), permafrost is thawing at accelerating rates. **Mechanism:** thawing exposes previously frozen organic matter to microbial decomposition: aerobic = CO_2; anaerobic (in waterlogged areas, thermokarst lakes) = CH_4 (80x more potent than CO_2 over 20 years). **Scale:** current permafrost emissions: ~1.5 Gt CO_2-equivalent/yr. Models project 92–260 Gt C release by 2100 under high-emissions scenarios — equivalent to decades of current global emissions. **Positive feedback:** warming → permafrost thaws → carbon released → more warming → more thawing. This feedback is largely irreversible on human timescales. **Observable impacts:** ground subsidence (buildings, roads, pipelines collapsing in Siberia), thermokarst lakes expanding and bubbling with methane, coastal erosion accelerating as frozen cliffs thaw. Russia's infrastructure damage from permafrost thaw could cost $100+ billion by 2050.

4.3 Energy Security

Energy security: The uninterrupted availability of energy sources at an affordable price (IEA). Encompasses reliable supply, affordable access, and environmental sustainability — the **energy trilemma**. No country achieves perfect energy security; all face trade-offs.

Global Energy Mix (2023)

Source	% Global Primary Energy	Key Facts
Oil	~31%	Transport fuel (~56% of oil demand), petrochemicals. OPEC+ controls ~40% of production. Largest producers: USA, Saudi Arabia, Russia.
Coal	~26%	Electricity generation, steel production. Cheapest but most polluting fossil fuel. China consumes ~54% of global coal.
Natural gas	~23%	"Transition fuel" (50% less CO_2 than coal). LNG trade growing rapidly. Russia, Qatar, USA are top exporters.
Hydroelectric	~7%	Largest renewable electricity source. China (Three Gorges: 22.5 GW), Brazil (Itaipu: 14 GW) are leaders.
Nuclear	~4%	Low-carbon but controversial. France: 70% of electricity from nuclear. China: 23 reactors under construction.
Wind & solar	~6%	Fastest-growing. Solar costs fell 90% since 2010. In 2023, solar became cheapest new electricity in most of world.
Bioenergy/other	~3%	Traditional biomass (cooking in LICs), modern biofuels, geothermal, tidal, wave.

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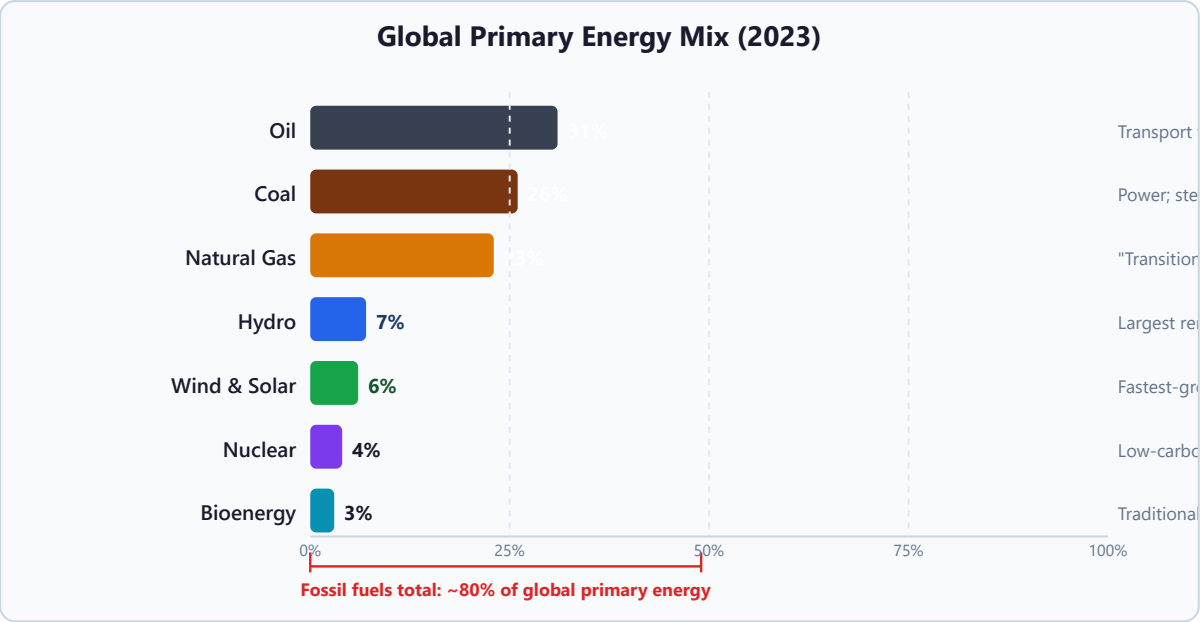

Fossil Fuel Dependency and Geopolitics

- **Uneven distribution** — fossil fuel reserves are concentrated: Middle East holds 48% of proven oil reserves; Russia has 24% of global gas reserves; the USA, China, and India hold 60% of coal reserves. This creates energy dependency and geopolitical leverage.
- **OPEC+ and oil prices** — the cartel of 23 oil-producing nations can manipulate global oil prices through production quotas. Oil price ranged from -$37/barrel (April 2020, COVID demand collapse) to $130/barrel (March 2022, Russia-Ukraine war).
- **Russia-Europe gas dependency** — before 2022, Russia supplied ~40% of Europe's gas via pipelines (Nord Stream 1 and 2, Yamal-Europe, TurkStream). Russia's invasion of Ukraine in 2022 triggered Europe's worst energy crisis since the 1970s: gas prices surged 400%, triggering a cost-of-living crisis. Europe rapidly diversified to LNG (from USA, Qatar, Norway), accelerated renewables, and reduced demand through efficiency measures.
- **Resource curse** — many resource-rich LICs suffer from the "paradox of plenty": oil wealth concentrates in ruling elites, fuels corruption, distorts economies (Dutch Disease), and can trigger conflict. Nigeria: world's 6th largest oil producer but 40% of population lives in poverty; Niger Delta oil extraction has caused severe environmental degradation.

The Energy Transition

- **Renewable energy growth** — global renewable capacity additions reached 507 GW in 2023 (50% increase on 2022). Solar alone added 420 GW. Renewables are now cheaper than fossil fuels for new electricity generation in most countries.

- **Challenges of transition** — intermittency (wind and solar vary with weather/season; requires storage), grid infrastructure upgrades, critical mineral supply (lithium, cobalt, rare earths — see below), stranded assets (fossil fuel infrastructure becoming economically unviable), social impacts (job losses in coal communities, energy poverty during transition).
- **Critical minerals** — EVs, wind turbines, solar panels, and batteries require vastly more minerals than fossil fuel technologies. Global lithium demand projected to increase 42x by 2040. Supply concentrated: DRC (70% of cobalt), China (60% of lithium processing, 90% of rare earths), Chile (42% of lithium reserves). Creates new geopolitical dependencies.
- **Carbon capture and storage (CCS)** — technology to capture CO_2 from industrial sources and store it in geological formations. Sleipner (Norway) has stored ~1 Mt CO_2/yr since 1996. Currently expensive (~$50–100/tonne CO_2) and limited in scale (total global CCS capacity: ~45 Mt CO_2/yr, compared to ~37 Gt CO_2/yr emissions). Seen by some as necessary for hard-to-decarbonise industries (cement, steel) but criticised by others as prolonging fossil fuel dependence.
- **Hydrogen** — "green hydrogen" (produced by electrolysis using renewable electricity) could decarbonise heavy industry, shipping, and aviation. Currently expensive and energy-intensive; costs falling rapidly.

Case Study: UK Energy Security and Transition

Challenge: North Sea oil/gas peaked ~1999; UK became net energy importer in 2004. Now imports ~50% of gas (Norway, LNG) and ~80% of oil. 2022 energy crisis (Russia-Ukraine) exposed vulnerability: wholesale gas prices surged 400%; government spent £37 billion on household energy support. **Renewable progress:** renewables generated ~42% of UK electricity in 2023 (wind 27%, solar 4%, biomass 6%, hydro 2%). UK has 14 GW offshore wind (world's 2nd largest after China), targeting 50 GW by 2030. **Controversies:** new North Sea oil licences (Rosebank field approved 2023 despite climate protests); nuclear new-build (Hinkley Point C: delayed to 2029, cost £33+ billion, originally £18 billion); fracking banned since 2019 after induced seismicity; onshore wind effectively banned in England 2015–2023 (planning restrictions partially lifted). **Energy poverty:** 3.2 million UK households in fuel poverty. **Net zero target:** UK legally committed to net zero emissions by 2050 (Climate Change Act 2008, amended 2019). Requires near-complete decarbonisation of electricity (by 2035), transport (petrol/diesel car sales banned from 2035), heating (heat pump rollout), and industry.

Case Study: China's Energy Dilemma

China is the world's largest energy consumer (~27% of global total) and CO_2 emitter (~31% of global

emissions). Coal provides ~55% of primary energy and ~60% of electricity. However, China is simultaneously the world's largest investor in clean energy: ~$758 billion in 2023, more than rest of world combined. China manufactures 80% of global solar panels, 60% of wind turbines, and dominates EV production (60% of global EV sales in 2023). China added 217 GW of solar capacity in 2023 alone — more than any other country's total installed solar capacity. **Tension:** rapid industrialisation demands energy security and affordability (justifying continued coal expansion — China approved 106 GW of new coal plants in 2022); severe air pollution (1.2 million premature deaths/yr) and climate commitments (carbon peak by 2030, net zero by 2060) drive renewable transition. China's approach reflects the energy trilemma at the largest scale: balancing economic growth, energy access for 1.4 billion people, and environmental sustainability. China's Belt and Road Initiative exports both coal power plants AND renewable technology to developing countries, reflecting this internal contradiction.

4.4 Managing Carbon Emissions

International Agreements

- **UNFCCC (1992)** — Framework Convention on Climate Change. Established the principle of "common but differentiated responsibilities" (developed countries bear greater responsibility for emissions reductions).
- **Kyoto Protocol (1997)** — legally binding emission reduction targets for developed countries only. USA never ratified. Limited effectiveness but established carbon trading mechanisms.
- **Paris Agreement (2015)** — 196 nations agreed to limit warming to well below 2°C, aiming for 1.5°C. Each country sets its own Nationally Determined Contribution (NDC). Not legally binding for individual targets. Current NDCs are insufficient: projected to result in 2.5–2.9°C warming.
- **COP28 Dubai (2023)** — first agreement to "transition away from fossil fuels" (landmark language, though weaker than "phase out"). Pledged tripling of renewable energy capacity by 2030.

Carbon Management Strategies

- **Carbon taxation** — pricing carbon emissions to incentivise reduction. Sweden's carbon tax ($130/tonne, highest in world) has reduced emissions 27% since 1991 while GDP grew 80%.
- **Emissions trading (cap and trade)** — EU Emissions Trading System (EU ETS) caps total emissions and allows companies to trade permits. Carbon price reached €100/tonne in 2023. Effective but complex

and subject to political lobbying.

- **Afforestation and reforestation** — planting trees to absorb CO_2. China's "Great Green Wall" aims to plant 350,000 km^2 of forest by 2050. Limitations: takes decades for trees to reach maximum sequestration; land competition with food production; monoculture plantations have lower biodiversity and carbon storage than natural forests.
- **Peatland restoration** — rewetting drained peatlands to restore carbon sink function. UK has 3 million hectares of peatland storing 3.2 billion tonnes C; 80% degraded. UK Peat Action Plan aims to restore 35,000 hectares by 2025.
- **Blue carbon** — protecting coastal ecosystems (mangroves, salt marshes, seagrass) that sequester carbon at 2–4x the rate of tropical forests per hectare. Globally, mangrove loss releases ~150 Mt CO_2/yr.
- **Carbon capture and storage (CCS)** — capturing CO_2 at source and storing underground. Currently limited scale (45 Mt CO_2/yr capacity globally vs 37 Gt CO_2/yr emissions). Expensive but potentially necessary for cement, steel, and gas-fired power.
- **Direct Air Capture (DAC)** — machines that extract CO_2 directly from ambient air. Climeworks' Orca plant (Iceland) captures 4,000 tonnes CO_2/yr. Currently extremely expensive ($400–600/tonne) and energy-intensive.

Exam Tip: Edexcel links the carbon cycle directly to energy security. Always show how fossil fuel burning disrupts the carbon cycle (transferring carbon from slow geological stores to the fast atmospheric cycle) and how energy security strategies have carbon cycle implications. Renewables reduce atmospheric carbon additions; CCS attempts to return carbon to geological storage; nuclear is low-carbon but creates waste storage challenges. This integrated approach earns top marks.

Exam Tip: When evaluating carbon management strategies, consider: scale (can it make a significant difference to the 37 Gt CO_2/yr problem?), cost (who pays?), time (how quickly does it work?), co-benefits (biodiversity, jobs, health), trade-offs (land use competition, resource requirements), and equity (does it benefit or burden developing countries?). No single strategy is sufficient; a portfolio approach is needed.

Exam Tip: For Edexcel, always distinguish between the fast and slow carbon cycles. Human fossil fuel combustion is unique because it transfers carbon from the slow geological cycle (where it was sequestered over millions of years) to the fast atmospheric cycle (where it causes warming) at a rate the natural system cannot absorb. This framing demonstrates sophisticated understanding of carbon cycle dynamics.

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